

# Narrative Web Visualization

Design and develop a hybrid narrative web visualization.

Choose a theme of interest and search relevant dataset(s).

- Convey the message utilizing a scrollytelling approach, where the story unfolds progressively as the user scrolls.
- Combine author-driven and reader-driven visualization into a single, engaging web based experience.

Project submission is due on 10<sup>th</sup> May, 2026. All students are required to make the submission.

## Project Workflow

**Step 1:** Extensive Data Analysis and Exploration: for your chosen theme, conduct data exploration & analysis to first, identify a question you would like to explore as part of your final project; and, then, second systematically delve into the dataset(s) so as to thoroughly address this question. What aspects of the data reveal the most interesting discoveries or stories? Do not feel obligated to try to convey everything about the data: focus on a compelling subset.

Conduct sufficient exploration and analysis such that, at the end of this task, you have a clear idea of what the takeaway insights from your final project be for your target audience, and how you will use the data to support your arguments. Moreover, you should have this data ready for use in building out your final project including cleaning, shaping, and transforming it as necessary.

**Step 2:** Storyboarding through sequence of sketches: they should communicate the overall narrative structure, visual layout, and interactive flow of your project by providing a step-by-step walkthrough of its main features.

They may guide regarding: What is the arc of your narrative, and what are the main story beats? How many different visualizations will you have, and how do they correspond to the narrative beats? How should the visualizations and narrative pieces be laid out, and how should the users navigate between them?

Then go one step deeper and consider the specific design and capabilities of the visualizations—for example, what kinds of interactive or animation techniques might you use to make things more engaging, and what visualizations remain static? How should you use highlights, annotations, or other narrative features intended to draw attention to particular items of interest and provide additional context?

**Step 3:** Visual design study: Whereas in the storyboarding focus on structure, layout, and flow, a visual design study instead helps you establish the “look and feel” of your final project—namely, e.g. fonts and how do they also help communicate the takeaway message(s) to your intended audience? Look through a diverse range of visual media (including web, screenshots, newsprint, graphic novels, etc.)

**Step 4:** Implementation: A bug-free, polished, deployed implementation of your final project . Make sure your ultimate project includes the following at the bottom:

- The names of all members of your team
- Acknowledgements for all appropriate sources (both for your datasets, and also any example visualizations, code, etc. that you might have adapted for your purposes).

### Creativity expectations

Evaluation is also on creativity. It is encouraged to:

- Choose unique and impactful topics
- Design engaging interactions
- Use animations and transitions effectively
- Combine multiple visualization techniques

### Presentation:

Present your project in a logical narrative flow:

1. Introduce yourself and team member and mention the reg numbers.
2. Present research Question & Motivation
3. Live demo of narrative visualization

Presentation should reflect clear storytelling. Each member should present.

### Deliverables:

Project requires the submission of following deliverables on MS Teams.

- A properly formatted report covering the outcome of Step 1, 2, and 3. These are also mentioned in the rubric below. Name the deliverable as “ProjectReport\_2023xxx\_2023xxx.pdf”
- The publicly accessible URL of the final project covering narrative visualization. The project may be made accessible through github pages.

## Rubric:

| Component                                 | Excellent                                                                                                                                                                                                                                                                                                                                                                     | Satisfactory                                                                                                                                                                                                                                                                                                                            | Needs Improvement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Research Question &amp; Motivation</b> | <p>Your data analysis has yielded a compelling question (i.e., one without a straightforward or intuitive answer, and thus ideal to unpack over a data story). You went above and beyond in finding extra materials to help motivate the question, including via additional background research that richly illustrates the impact/implications on housing affordability.</p> | <p>A reasonably interesting question is posed, and is relatively well-grounded in the materials provided. Your submission begins to demonstrate an understanding of why this issue is important and worth addressing. Further analysis or background research would help to nuance the question and unpack its impact/implications.</p> | <p>You can lose points for any of the following:<br/>           (a) the question is difficult to understand, vague, or posed too broadly to be addressable in a final project;<br/>           (b) conversely, the question may be too narrowly posed, or offer too straightforward or intuitive an answer/explanation such that it would be difficult to produce a compelling data story; (c) motivation or cited evidence is missing, communicated poorly, or is mismatched to the question</p> |
| <b>Data Analysis Results</b>              | <p>A deep and thorough analytical treatment of your subtheme and research question has yielded a rich and insightful set of results that leave practically no stone unturned. This thoroughness may emerge from a variety of ways including but not</p>                                                                                                                       | <p>Sufficient analysis has been conducted to yield results that point to a potentially compelling set of insights. However, following some analytic threads down a few additional steps of depth would have yielded a richness and texture to the details (e.g.,</p>                                                                    | <p>The analysis process appears to have been conducted in a relatively ad hoc or piecemeal fashion that has yielded relatively shallow results. Several missed opportunities to ask follow-up questions or follow a thread in more detail. Incorrect analysis</p>                                                                                                                                                                                                                                |

| Component           | Excellent                                                                                                                                                                                                                                                                                                                                                                 | Satisfactory                                                                                                                                                                                                                                                                                                                                | Needs Improvement                                                                                                                                                                                                                                                                              |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | <p>limited to:</p> <p>(a) interrogating the data from multiple different angles;</p> <p>(b) advanced data transformation;</p> <p>(c) synthesizing across multiple datasets or integrating external datasets.</p> <p>(Note: your wireframed narrative is still crisp; analysis thoroughness is demonstrated through supplementary result slides in your presentation).</p> | <p>uncovering some nuance, surprise, or just lending a sense of “completeness” to the emerging story).</p>                                                                                                                                                                                                                                  | <p>may have been conducted in parts.</p>                                                                                                                                                                                                                                                       |
| Storyboarding       | <p>The storyboards richly depict the narrative flow of your final project. It is easy to imagine how your intended audience should experience your work, and the impact it should have on them. The should make clear what the major pieces of analysis, design, and implementation work your final project should encompass.</p>                                         | <p>Your storyboards do a good job of describing your final project. But, there are occasional missing details that make it difficult to robustly understand how your intended audience will experience your final project. These missing details also somewhat hinder your ability to plan out the rest of your final project timeline.</p> | <p>Your storyboards are very rough, making it difficult to understand your final project’s narrative structure, or how your intended audience will experience them. They are not especially helpful in planning design and implementation details, or dividing labor between team members.</p> |
| Visual Design Study | <p>The design study explores a diverse range of visual</p>                                                                                                                                                                                                                                                                                                                | <p>The design study offers some nice inspiration for</p>                                                                                                                                                                                                                                                                                    | <p>The design study is missing or incomplete</p>                                                                                                                                                                                                                                               |

| Component              | Excellent                                                                                                                                                                                                                                                                                        | Satisfactory                                                                                                                                                                                                                                                                                                               | Needs Improvement                                                                                                                                                                                                             |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Interaction Techniques | media and representation strategies. Typographic inspiration are well-chosen for the intended audience and help to reinforce the tone, information, and takeaway message that needs to be communicated.                                                                                          | your project's look and feel, but could have drawn from a more diverse range of sources. It is a little difficult to see how certain design choices connect to your intended audience or narrative purpose.                                                                                                                | (e.g., only focuses on color or typography, or draws from a very limited pool). Specific inspiration is not identified or is poorly communicated.                                                                             |
|                        | Compelling interaction techniques actively engage in the viewer in a process of meaningful exploration, learning, storytelling, or counterfactual reasoning. The techniques enable more effective discovery of interesting trends, patterns, or outliers. No notable bugs or performance issues. | Interaction techniques do a nice job of engaging the reader by helping to deepen the takeaways conveyed by the static visualization. But, some insights that are key to the narrative are unnecessarily "hidden" behind interactivity which occasionally yields a frustrating experience. Some bugs or performance issues. | Interaction techniques are overly simplistic and/or do not add much to a reader's experience beyond the static visualization. Major bugs and/or performance issues hinder the overall user experience.                        |
|                        | Your presentation is crisp, perhaps even exciting. Narration is clear, flows smoothly, and is easy for the audience to follow. Your presentation fits within the allotted 5 minutes.                                                                                                             | The presentation is clear, and we understand the goals of your project. However, some issues format, narration, or design hinder easy comprehension. Your presentation fits within the allotted 5 minutes.                                                                                                                 | Your presentation is difficult to follow and/or doesn't hold together. You may have gone over the time limit. Narration rambles and/or is difficult to follow. Visualization are text-heavy, making poor or inadequate use of |
| In-Class Presentation  |                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                               |

| Component | Excellent | Satisfactory | Needs Improvement                                                                 |
|-----------|-----------|--------------|-----------------------------------------------------------------------------------|
|           |           |              | images or visual media.<br>No members of the team showed up for the presentation. |

Evaluation is not limited to above aspects. Projects may also be evaluated on overall creativity, visualizations, quality and qualitative aspects. All operational guidelines such as project and group submission in time, naming file properly, and others must be followed. Marks may be deducted if these guidelines are not followed.

### Reference

<https://vis-society.github.io/>